GATE DA · DBMS solutions guide

Hints, worked steps and final checks for P01–P40, followed by answer summaries and reasoning for all four mixed tests. This guide solves the original problems in the companion practice book. Worked historical-question explanations remain in the notes next to their source links.

Chapter 1 · Solutions

Try each question before reading this guide. The hint gives the first move. The worked steps show why the answer follows. Letters I, N and E in the verification log stand for ID, Name and Email.

P01 · Keys and counts

Hint. Start with the declared identifiers. Add other columns to make superkeys.

The candidate keys are {ID} and {Email}. Their supersets are {ID}, {Email}, {ID,Name}, {ID,Email}, {Name,Email}, and {ID,Name,Email}. That is six superkeys. {Name} is not guaranteed to be unique in every permitted instance.

The degree is 3 and the cardinality is 4. After a fifth row, the cardinality becomes 5. The schema still has three attributes, so its degree remains 3.

Check. Each listed superkey contains at least one candidate key. A current coincidence cannot establish a schema rule.

P02 · Foreign-key changes

Hint. Ask whether any remaining child points to a missing parent.

A and B are safe for the named foreign key. Adding parent 3 does not remove any referenced value. Deleting child 10 removes a reference.

C fails because parent 9 does not exist. D fails because child 11 still references parent 2. Under ON DELETE SET NULL, D succeeds: child 11's reference changes to NULL. The final child rows are (10,1), (11,NULL), (12,NULL).

Check. SET NULL also requires a nullable child column. These conclusions do not guarantee compliance with unrelated constraints.

P03 · Join counts

Hint. Count the matches of each purchase separately.

The references 1,1,2,NULL have match counts 1,1,1,0. The inner join has 3 rows. The left join preserves the unmatched purchase and therefore has 4 rows.

After adding reference 3, the match counts are 1,1,1,0,1. The inner join has 4 rows and the left join has 5. Customer 1 still occurs twice in the inner join. A join returns matching pairs, not one row per customer.

Check. Uniqueness on the parent prevents a purchase from matching several parents. It does not prevent several purchases from matching the same parent.

Chapter 2 · Solutions

P04 · Trace the expression

Hint. Match Q first. Do not discard duplicates before projecting.

The natural join has five rows:

P	Q	R
1	a	9
1	a	10
2	a	9
2	a	10
3	b	10

R=10 keeps three rows. Their projected Q values are a,a,b, which form the set {a,b}. The final cardinality is 2. After removing (b,10), the first four joined rows remain. Selection leaves (1,a,10),(2,a,10), and projection returns {a}, with cardinality 1. The product sizes are 3×3=9 and 3×2=6.

Check. A join matches rows. A product makes every pair. Set projection removes duplicate projected tuples.

P05 · Count matching pairs

Hint. Group each side by the joining value.

Value 1 contributes 2×3=6 pairs. Value 2 contributes 1×1=1. Value 3 has no match. Total: 7. After making the S join column unique, value 1 contributes 2 and value 2 contributes 1, giving 3.

The bound |R| requires uniqueness on S's joining value. It does not apply to the original S, where each value-1 row of R has three partners.

P06 · Every required course

Hint. Make one set of courses per candidate student.

Student 1 has {M,P}; student 2 has {M}. The nonempty requirement is met only by 1. When Required is empty, both candidates from Enrolment pass, so classical division returns {1,2}.

A query starting with Student has candidate domain {1,2,3}. No candidate can miss a member of an empty Required set. It returns {1,2,3}, including student 3. The outputs differ because their candidate domains differ.

P07 · Disprove the claims

Hint. One name and an empty relation are enough.

For (a), choose Team={Asha}, Defender={Asha}, Forward=empty. The union covers Asha, so the first difference is empty. The intersection is empty, so the second difference contains Asha.

For (b), choose Team={Asha} and both player sets empty. Their union minus Team is empty. But Team minus their union contains Asha, so Team coverage fails.

Check. T minus U empty means T⊆U. U minus T empty means U⊆T. They are different directions.

P08 · Output rows versus witnesses

Hint. Build the witness list, then count distinct qualifying output tuples.

For >1, row (1,a) has witnesses (1,10),(1,20), and (2,b) has (2,30). Thus there are 3 witness pairs but 2 output tuples. Row (3,c) has no qualifying witness.

For ≥1, (3,1) also becomes a witness. There are now 4 witness pairs and 3 output tuples. The additional z condition is always satisfied by z=x. It changes neither result.

Check. Distinct variable names do not impose distinct values. An explicit inequality would be needed.

Chapter 3 · Solutions

P09 · Rows versus customers

Hint. One purchase produces one joined match.

The joined names are Asha,Asha,Ravi: 3 rows. DISTINCT returns Asha,Ravi: 2 rows. EXISTS also returns 2 customer rows because each customer has a purchase.

Adding (1,40) gives Asha,Asha,Asha,Ravi: 4 joined rows. DISTINCT and EXISTS still return 2. EXISTS tests whether a witness exists; it does not output one row per witness.

P10 · Missing values and empty input

Hint. List the non-null inputs before calculating the aggregate.

Quantity	All rows	WHERE Value>100
COUNT(*)	4	0
COUNT(Value)	3	0
COUNT(DISTINCT Value)	2	0
SUM(Value)	40	NULL
AVG(Value)	40/3	NULL

The ungrouped aggregate returns one result row even on empty input. GROUP BY Value on that input returns zero groups and zero rows.

Check. NULL is ignored by the value aggregates; it is not replaced by zero. SUM(empty) is not the same SQL result as summing observed zero values.

P11 · Filter then group

Hint. Apply amount>8 before making groups.

Initially WHERE leaves (A,10),(A,20). Their A group sums to 30 and passes HAVING. The output is (A,30).

After adding (B,30), WHERE leaves those two A rows and (B,30). Both groups sum to 30 and pass. The output is (A,30),(B,30). B's original 7 and NULL are excluded before grouping.

Before WHERE, B has COUNT(*)=2 and COUNT(amount)=1. After the added row those become 3 and 2. They disagree because one row has NULL amount.

P12 · Correlated comparisons

Hint. Count strict inequalities. Do not assign a shortcut rank to tied rows.

Balance	Above	Below	Keep?
5, first row	3	0	yes
5, second row	3	0	yes
10	2	2	yes
15	1	3	no
18	0	4	no

The count is 3. After replacing 15 by 10, each 10 row has 1 strictly above and 2 strictly below, so neither passes. Each 5 row still passes. The count becomes 2.

P13 · NOT IN and NOT EXISTS

Hint. Evaluate an unmatched ID against every Rejected value.

For IDs 1 and 3, IN returns UNKNOWN because of the NULL member and no true equality. NOT IN remains UNKNOWN, so WHERE rejects them. ID 2 is explicitly present and is also rejected by NOT IN. Output: empty.

NOT EXISTS finds a real equality match only for ID 2. It returns 1 and 3. After removing NULL, both queries return 1 and 3.

5>ANY(empty) is FALSE. 5>ALL(empty) is TRUE: there is no element that violates the comparison. This empty-set convention follows the quantifiers, not a MAX(empty) substitution.

P14 · Preserving rows and ordered pairs

Hint. ON forms the join. WHERE can then remove its preserved rows.

With the amount condition in ON, the rows are (Asha,20),(Ravi,NULL). With it in WHERE, only (Asha,20) remains. NULL>15 is UNKNOWN, so Ravi does not pass.

For self-join values 1,1,2, each of the two value-1 rows matches the one value-2 row under <. There are 2 ordered pairs. Under ≤, the two value-1 rows match both value-1 rows and the value-2 row: 2×3=6 pairs. The value-2 row matches itself: 1 more. Total: 7.

Check. A self join includes the same underlying row on both sides unless the predicate rules it out.

Chapter 4 · Solutions

P15 · Apply whole determinants

Hint. Start with the listed attributes, then fire every available rule.

(VW)+ starts {V,W}. V→X adds X; WX→Y and WX→Z add Y,Z. Final closure: VWXYZ. U is missing, so VW→U does not follow. U+ adds V,W, then X,Y,Z, and is UVWXYZ.

Without V→X, (VW)+ stays VW. X is missing, so neither WX rule can fire. U+ becomes UVW. Having W alone is not enough for a WX determinant.

P16 · Find every minimal identifier

Hint. C cannot be obtained from another attribute.

Every key contains C. C+ is CD. Either A or B supplies the missing pair A,B, giving keys AC and BC. Removing either member of either key loses full determination. Prime attributes are A,B,C.

After removing B→A, A and C are both compulsory: neither occurs on a right side. AC is the sole candidate key and A,C are prime. BC+ is BCD and misses A.

P17 · Count the union of supersets

Hint. Count each key's supersets, then subtract their overlap.

On ABCD, each two-attribute key has 2^2=4 supersets. Sets containing both keys must contain ABC, leaving D optional: 2 overlaps. Count: 4+4−2=6.

With A→E, both AC and BC still determine everything. On five attributes each has 2^3=8 supersets. Their overlap has ABC compulsory and D,E optional, giving 4. Count: 8+8−4=12.

With an unconstrained E, every key must include E. Keys become ACE and BCE. Each has four supersets and their overlap has two. Count returns to 6. More columns do not necessarily mean more optional choices.

P18 · Simplify without changing the rules

Hint. Test removals by closure after removing the rule being tested.

Split A→BC into A→B and A→C. B is extraneous in AB→D because A already obtains B, so replace it by A→D. A→C is redundant because A→B→C remains. The resulting G is {A→B,B→C,A→D}.

Each original rule follows from G, and each G rule follows from F. Every right side is a singleton. Removing any G rule loses its consequence, and no left side can be shortened. Thus G is a minimal cover.

Without A→D, A+ is ABC rather than ABCD. AB→D no longer follows, so the smaller G is not equivalent to F.

P19 · Keys before normal forms

Hint. Prime means belongs to at least one candidate key.

With AB→C,C→B, the keys are AB and AC. All attributes are prime. C+ is BC, so C is not a superkey. C→B violates BCNF, but B is prime and satisfies the 3NF exception. Highest form: 3NF.

With A→B,B→C, the only key is A. B→C has a non-superkey determinant and a nonprime dependent, so it violates 3NF. There is no partial dependency on a proper nonempty part of the one-attribute key. Highest form: 2NF.

P20 · Can the join invent rows?

Hint. For a binary split, close its common attributes.

AB/AC has common A. A+ includes AB, so it is lossless. AB/BC has common B, whose closure is B. It determines neither AB nor BC, so this split is not guaranteed lossless.

Original legal rows (a1,b1,c1),(a2,b1,c2) satisfy A→B. Their projections join into four rows; (a1,b1,c2),(a2,b1,c1) are extra. This proves the failure.

After adding B→C, B+ includes BC. The AB/BC split is now lossless. The old counterexample violates the added rule and can no longer be used.

P21 · Chase the three-way split

Hint. A present attribute gets its shared a symbol. An absent attribute gets a fresh b symbol.

Component	A	B	C	D
AB	aA	aB	b0C	b0D
BC	b1A	aB	aC	b1D
AD	aA	b2B	b2C	aD

A→B merges b2B with aB because AB and AD agree on A. B→C then makes the three C entries agree with the distinguished aC.

Component	A	B	C	D
AB	aA	aB	aC	b0D
BC	b1A	aB	aC	b1D
AD	aA	aB	aC	aD

AD is entirely distinguished, so the split is lossless. Without B→C, AD ends (aA,aB,b2C,aD). No row is entirely distinguished, and no further rule applies. The split is not guaranteed lossless.

Check. This is the FD chase, not a one-time comparison of intersections.

P22 · Preserve constraints locally

Hint. Project implied dependencies, not just the originally written ones.

AB/AC has local rules A→B and A→C. Their union gives B+={B}, so B→C is not preserved. The split is nevertheless lossless because the common A determines AB.

AB/BC has local A→B and B→C. Both original rules are enforceable without joining. Its common B determines BC, so it is also lossless. Here both properties hold.

P23 · Choose the guarantee you need

Hint. A BCNF split can be lossless without preserving every dependency.

Using C→B yields BC and AC. C is a key of BC; AC has no nontrivial projected violation. Both components are BCNF. Their common C determines BC, so the split is lossless.

Their local rules only provide C→B. Starting AB under these local rules does not obtain C, so AB→C is not preserved.

3NF synthesis produces ABC from AB→C and BC from C→B. BC is contained in ABC and can be removed. ABC contains key AB, so no extra key relation is needed. Keeping ABC is acceptable because the target is 3NF: its prime-attribute exception permits C→B. It preserves all dependencies and has no information-loss issue.

Chapter 5 · Solutions

P24 · Count pairs, then check feasibility

Hint. Total means each entity has at least one partner. Many-to-one adds an upper bound on A's partners.

In many-to-many, at least 3 pairs cover the 3 A entities. The pairs (a1,b1),(a2,b1),(a3,b2) also cover both B entities, attaining the bound. All six possible A×B pairs attain the maximum 6.

In many-to-one, each A has at most one B. Total participation gives at least one, so exactly one. There are exactly 3 pairs. The same minimum example covers both B entities.

With only one A, there can be only one pair, but both B entities require a partner. The conditions are infeasible. Do not report a numerical minimum for an impossible instance.

P25 · Keep each fact once

Hint. Identify independent objects, repeated attributes and relationship facts separately.

Table	Primary key	Reference or purpose
Department(DeptID,Name)	DeptID	one department
Student(StudentID,Name,DeptID)	StudentID	DeptID references Department; NOT NULL
Course(CourseID,Title)	CourseID	one course
Phone(StudentID,Phone)	StudentID,Phone	owner references Student
Enrolment(StudentID,CourseID)	StudentID,CourseID	references both entities
Dependent(StudentID,DependentName)	StudentID,DependentName	owner references Student

There are 6 tables under the stated facts. DependentName alone need not be unique across students. A nullable Phone column in Student can represent at most one phone, so the Phone table can disappear in (b). The count becomes 5.

Check. A several-phone attribute cannot be stored as a repeating group in the original atomic-value Student schema.

P26 · A changed upper bound changes the mapping

Hint. A many-to-many pair is an independent fact; a many-to-one link can be a reference.

The four tables are E1, E2, E2Value(E2ID,Value), and Link(E1ID,E2ID). The multivalued table preserves independent values; the link table preserves all permitted pairs. Both-side total participation does not remove either need.

With many-to-one from E1 to E2, put a NOT NULL E2ID reference in E1 and remove Link. The minimum becomes 3. E2ID need not be unique in E1 because several E1 entities may share one E2.

Chapter 6 · Solutions

P27 · Pack whole records

Hint. A block's leftover bytes cannot hold part of the next unspanned record.

Records per block=floor(1024/100)=10. Required blocks=ceil(101/10)=11. The shortcut ceil(10100/1024)=10 treats unused space as available across block boundaries, which is not allowed.

With headers, usable bytes=1024−50=974. Capacity=floor(974/100)=9. Required blocks=ceil(101/9)=12. Check: 11 blocks hold only 99 records.

P28 · Count entries before levels

Hint. Sparse anchors count data blocks; dense entries count records.

Data capacity is 10 records per block, so the file has 1000 blocks. An index block holds 1000/20=50 entries.

Layout	Lowest-level entries	Index blocks, bottom to top	Index levels	Lookup including data
sparse	1000	20,1	2	3 I/Os
dense	10,000	200,4,1	3	4 I/Os

The top has to fit in one block. Count one page at each index level, then the requested data page.

P29 · Write both byte budgets

Hint. An internal node has one more pointer than separator.

$4p+8(p-1)\le 1024$ gives $p\le 86$. At $p=86$ the usage is 1024; $p=87$ does not fit. Leaf entries satisfy $20q+4\le 1024$, giving $q=51$.

After doubling child pointers, $8p+8(p-1)\le 1024$ gives $16p\le 1032$, so $p=64$. The leaf record pointer and link did not change, so leaf capacity remains 51.

P30 · Build from the leaves

Hint. Round up at every parent layer.

10,000 records need 200 leaves, then 2 parents, then 1 root. Height: 3 levels or 2 edges. 5000 records need 100 leaves and one root. Height: 2 levels or 1 edge.

For exactly 3 levels, the minimum is 2 root children ×50 leaves per internal child ×25 records per leaf=2500. The maximum is 100×100×50=500,000. Root occupancy and leaf capacity are different factors.

P31 · Repair each overflow or underflow

Hint. Copy a leaf separator. Promote an internal separator.

Initial tree:

```
      [7]
     /   \
[1,4]  [7,10]
```

Insert 2. Leaf [1,2,4] splits into [1,2] and [4]; copy 4 to the root:

```
      [4,7]
     /  |  \
[1,2] [4] [7,10]
```

Insert 3. Leaf [1,2,3] splits into [1,2] and [3]. The root would be [3,4,7], with four children, so it overflows. Promote 4:

```
        [4]
       /   \
     [3]   [7]
    /  \   /  \
[1,2] [3] [4] [7,10]
```

Delete 10. The last leaf becomes [7] and meets minimum occupancy. Delete 7. Its leaf becomes empty; [4] cannot lend, so merge them. The right internal node has only one child and underflows. The left internal node has only its minimum two children, so it cannot lend. Merge the internal nodes and collapse the root:

```
      [3,4]
     /  |  \
[1,2] [3] [4]
```

Check. All leaves stay at one depth. Each internal separator is the minimum of the right subtree. No record appears twice because of an internal separator.

P32 · Match the predicate and count pages

Hint. Hashing gives equality buckets; B+ order gives ranges.

Rating>3.4 uses B+. Genre='Comedy' and exact MovieID can use either standard hashing or B+ lookup.

A 3-level path with root cached reads two index pages. Add one data fetch: 3 I/Os. A covering lookup needs only 2.

For the range, the path's two reads already include the first leaf. Add the other four leaves and six data pages: 2+4+6=12 I/Os. Counting all five leaves again would double-count the first one.

Chapter 7 · Solutions

P33 · Grain before aggregation

Hint. Count each fact row's join partners.

The grain is one sale line. The star has one fact and three dimension tables: 4. Splitting category information into its own dimension relation gives 5 tables and a snowflaked product dimension.

The true sale sum is 100+50=150. A has two tag partners and B has one, so the joined amounts are 100,100,50, summing to 250. The join changes the grain to sale-line×tag. To sum sales, retain one contribution per sale line or use an existence filter when tags only determine eligibility.

P34 · Describe the actual change

Country→State is drill-down. City→State is roll-up. Fixing one Country value is a slice. Restricting multiple dimensions is a dice. Swapping axes is a pivot.

A larger output count does not prove drill-down. Extending a report from one month to two months can add rows while keeping its grouping levels unchanged. The levels, not the count alone, decide drill-down or roll-up.

P35 · Three quantities that look similar

Hint. Count observed groups within each grouping set.

Grouping set	Original groups	After adding (B,US,40)
Product,Country	3	4
Product	2	2
Country	2	2
no dimensions / grand total	1	1
Total output rows	8	9

There are 2^2=4 grouping sets in both cases. The first dataset is missing one detail combination, so the complete-product formula 3×3=9 does not apply to it.

For independent hierarchies, choosing one level per dimension gives 4×3×4=48 cuboids. Those counts already include ALL; adding it again would overcount.

P36 · Carry sufficient summaries

Hint. Recover totals, then divide by the total count.

The group sums are 2×100=200 and 8×50=400. Overall mean=(200+400)/(2+8)=60. Carry sum and count. AVG is algebraic; averaging the two subgroup averages without weights would give the wrong value 75.

The customer union is {1,2,3}, so the distinct count is 3. Adding subgroup counts gives 4 and counts customer 2 twice. Counts 2,2 could also describe disjoint sets with union size 4, proving that counts alone cannot determine the answer.

Balances can be summed across accounts at the same date. Adding balances across dates usually counts the same stock repeatedly and does not describe a balance at any date. Balance is semi-additive across those dimensions. This differs from the distributive/algebraic/holistic classification of computation.

Chapter 8 · Solutions

P37 · Identify the scale before using arithmetic

Hint. A numeric storage code is not automatically a quantitative measurement.

Department code is nominal; Low/Medium/High is ordinal; Celsius is interval; mass in kg is ratio. Mean department codes do not have a department interpretation. Ordinal labels have order without guaranteed equal gaps.

For x=60, min-max to [0,1] is (60−20)/(80−20)=2/3. To [-1,1], use −1+2×(2/3)=1/3. Z-score is (60−50)/10=1.

Decimal scaling needs 80/10^j<1. j=1 fails; j=2 works. The value becomes 60/100=0.60.

If all observations are 7, max−min=0 and SD=0. Ordinary min-max and z-score formulas divide by zero. A handling rule must be supplied; the formulas do not decide it. Relational normalization is unrelated to this numerical scaling.

P38 · Follow the boundary convention

Hint. Sort and assign bins before calculating replacement values.

Method	Bin memberships	Bin means	Smoothed sequence
equal width	{0,1,2}; {3}; {8,9}	1; 3; 8.5	1,1,1,3,8.5,8.5
equal frequency	{0,1}; {2,3}; {8,9}	0.5; 2.5; 8.5	0.5,0.5,2.5,2.5,8.5,8.5

3 belongs in [3,6), not [0,3). 9 belongs in the final closed interval. Equal-frequency bins use two observations each, not equal numerical widths.

Bin labels produce categorical membership labels in place of these numerical means. Neither labels nor mean smoothing generally permit exact recovery of every original value.

P39 · Population weights differ from sample allocation

Hint. Weight stratum means by their population shares.

The population-mean estimate is (90/100)×10+(10/100)×30=12. The unweighted observed sample mean is (5×10+5×30)/10=20. Equal sampling from unequal strata does not give equal population weights.

Two ordered draws without replacement have 5 choices followed by 4: 20 outcomes. With replacement, both draws have 5 choices: 25 outcomes. Unordered subsets would be a different counting problem.

The first-stage unit in the stated cluster design is a village. Including all its households does not turn the design into independent random household draws.

P40 · Count overhead and demand reconstruction

Hint. Count the actual encoded representation, not only its payload.

Encoded size=600+100=700 bytes. The original/encoded ratio is 1000/700=10/7≈1.4286. Saving=(1000−700)/1000×100%=30%.

AAAABBCD has runs (A,4),(B,2),(C,1),(D,1). Four 16-bit pairs plus an 8-bit header cost 72 bits. The original eight 8-bit symbols cost 64 bits. The encoding expands by 8 bits, or 12.5%. Run-length encoding is not automatically shorter.

There are four two-bit inputs: 00,01,10,11. Only three strings are shorter than two bits: the empty string,0,1. Exact reconstruction requires different inputs to have distinguishable encodings. Four inputs cannot all fit into three distinct shorter encodings. Thus a lossless code cannot shorten every two-bit input.

Mixed tests · Solutions and repairs

Use this after attempting a test. An MCQ has one correct choice; an MSQ needs every correct choice and no incorrect choice. A NAT result must answer the requested count or value. These are original exercises, not official exam questions.

Test 1 · Answer summary

Question	Answer
T1Q01	B
T1Q02	2
T1Q03	1
T1Q04	C
T1Q05	A; C
T1Q06	4
T1Q07	32
T1Q08	10
T1Q09	2
T1Q10	A; D

T1Q01 · B

Every Payment has a non-null valid reference and BuyerID is unique in Buyer. Each payment therefore gets exactly one join partner. Repeated buyers do not make BuyerID unique in Payment, so A fails. A referenced buyer cannot always be deleted, so C fails. Current distinct names do not establish D.

Repair if missed: review R01, R02, R03 and redo its worked example before trying a fresh variation.

T1Q02 · 2

The joined rows are (1,2,9),(1,3,9),(2,3,9). All pass C=9. Projection A gives the set {1,2}, removing the repeated 1. Count 2, not 3 joined rows.

Repair if missed: review R04 and redo its worked example before trying a fresh variation.

T1Q03 · 1

COUNT(*)=4 and COUNT(v)=3. Their difference is 1, the number of NULL values here. The duplicate 10 still contributes to COUNT(v).

Repair if missed: review R10 and redo its worked example before trying a fresh variation.

T1Q04 · C

D is compulsory because no rule derives it. AD supplies B,C. CD supplies A and then B. Both are minimal. BD cannot get A or C. The complete key set is {AD,CD}; A and C alone miss D.

Repair if missed: review R16 and redo its worked example before trying a fresh variation.

T1Q05 · A; C

The common A determines AB, so A is true. Projected local rules include A→B and A→CD, but B cannot obtain C,D locally: B is false. Both components have A as key and are BCNF, so C is true. The original has B→CD with a non-superkey determinant, so D is false.

Repair if missed: review R19, R20, R22 and redo its worked example before trying a fresh variation.

T1Q06 · 4

Each of four projects has exactly one leader, so it contributes one pair: 4. Total engineer participation is feasible: distribute the four projects as 2,1,1 among the three engineers. It does not require all 12 possible pairs.

Repair if missed: review R24 and redo its worked example before trying a fresh variation.

T1Q07 · 32

$8p+8(p-1) \le 512$ gives $16p \le 520$. Thus $p \le 32.5$ and the maximum integer is 32. $p=33$ uses 520 bytes and does not fit.

Repair if missed: review R29 and redo its worked example before trying a fresh variation.

T1Q08 · 10

Detail groups:4. First-dimension totals:3. Second-dimension totals:2. Grand total:1. Total=10. Six possible detail pairs do not all occur, so a full-product count would be wrong.

Repair if missed: review R35 and redo its worked example before trying a fresh variation.

T1Q09 · 2

$z=(15-9)/3=2$. The value is two standard deviations above the mean; it is not 2 percent or a min-max value.

Repair if missed: review R37 and redo its worked example before trying a fresh variation.

T1Q10 · A; D

A is true because total mean=(sum of subgroup sums)/(sum of subgroup counts). B is false: sets {1,2},{2,3} have counts 2,2 but union count 3. C is false because daily balances are a stock repeated across dates. D is true: each of three ordinary dimensions is included or aggregated away, yielding $2^3=8$ grouping sets.

Repair if missed: review R35, R36 and redo its worked example before trying a fresh variation.

Test 2 · Answer summary

Question	Answer
T2Q01	B; C
T2Q02	1
T2Q03	0
T2Q04	1
T2Q05	B
T2Q06	5
T2Q07	3
T2Q08	B
T2Q09	5
T2Q10	40

T2Q01 · B; C

A is false because many children may share a parent. B is true: SQL primary keys are non-null. C is true: on ABC, A→BC and BC→A make both A and BC minimal keys even though their sizes differ. D confuses one current instance with all permitted instances.

Repair if missed: review R01, R02 and redo its worked example before trying a fresh variation.

T2Q02 · 1

ID 1 has both a and b. ID 2 misses b and ID 3 misses a. Only one candidate has no missing requirement.

Repair if missed: review R06 and redo its worked example before trying a fresh variation.

T2Q03 · 0

IDs 1 and 3 have UNKNOWN membership because of the NULL. ID 2 is present and fails NOT IN. WHERE retains none. A matching-ID NOT EXISTS query would be a different result.

Repair if missed: review R13 and redo its worked example before trying a fresh variation.

T2Q04 · 1

For 2, above=3 and below=0, so it passes. For either 4, above=1 and below=1; strict > fails. For 9, above=0 and below=3. Only the 2 row remains. Ties matter.

Repair if missed: review R12 and redo its worked example before trying a fresh variation.

T2Q05 · B

AB is the only candidate key. Neither A nor B alone determines a nonprime attribute, so there is no partial dependency: 2NF holds. C→D has a non-superkey determinant and nonprime dependent, violating 3NF and BCNF. Hence B, not A,C or D.

Repair if missed: review R19 and redo its worked example before trying a fresh variation.

T2Q06 · 5

Owner and Hobby need two strong-entity tables. Dependent needs its own owner-plus-partial-key table. Skill needs one multivalued table. The many-to-many Dependent-Hobby relationship needs a link table. Total:5. Weak identification does not eliminate the Dependent relation.

Repair if missed: review R25, R26 and redo its worked example before trying a fresh variation.

T2Q07 · 3

Each block holds floor(1000/90)=11 records. Data blocks=ceil(950/11)=87. Index capacity=50 entries per block, so the sparse level needs 2 blocks and its next level needs 1. Read the root, one lower index block and one data block:3 I/Os.

Repair if missed: review R27, R28 and redo its worked example before trying a fresh variation.

T2Q08 · B

Month to Year moves to a coarser level: roll-up. Drill-down is the reverse. No value restriction or axis exchange is described, so dice and pivot do not fit.

Repair if missed: review R34 and redo its worked example before trying a fresh variation.

T2Q09 · 5

Equal-frequency bins are {0,2} and {4,6}, with means 1 and 5. The original 4 belongs to the second bin and becomes 5. Equal-frequency concerns observation counts.

Repair if missed: review R38 and redo its worked example before trying a fresh variation.

T2Q10 · 40

Two runs need 2×16=32 bits; add the 16-bit header to get 48. Original size=10×8=80 bits. Saving=(80−48)/80×100%=40%. The header is part of the representation.

Repair if missed: review R40 and redo its worked example before trying a fresh variation.

Test 3 · Answer summary

Question	Answer
T3Q01	4
T3Q02	A; C
T3Q03	A; C
T3Q04	2
T3Q05	C
T3Q06	C
T3Q07	6
T3Q08	12
T3Q09	11
T3Q10	D

T3Q01 · 4

Customer 1 has no partner and contributes one NULL-extended row. Customer 2 has two purchases and contributes two rows. Customer 3 has no partner and contributes one. Total=4. The NULL-referencing purchase is not a Customer match and is not separately preserved in this direction.

Repair if missed: review R03, R14 and redo its worked example before trying a fresh variation.

T3Q02 · A; C

A is valid because S contains C and the same C predicate can be tested before the join. B can remove the shared matching attribute and turn a natural join into a product: take R={(1,2)}, S={(3,5)}; the original join is empty, but πA(R) joins with S without a shared B and gives a row. C is the set-projection rule. D is false: natural join checks every shared name.

Repair if missed: review R04, R07 and redo its worked example before trying a fresh variation.

T3Q03 · A; C

On a one-row NULL table, COUNT(v)=0 but COUNT(*)=1, so A is true and B false. SELECT NULL still returns a row; EXISTS is TRUE, so C is true. NULL=NULL is UNKNOWN rather than TRUE, so D is false.

Repair if missed: review R10, R13 and redo its worked example before trying a fresh variation.

T3Q04 · 2

Remove A→C because A→B→C derives it. The remaining two rules are necessary: removing A→B loses B from A+, and removing B→C loses C from B+. The minimal singleton-right-side cover has 2 dependencies.

Repair if missed: review R18 and redo its worked example before trying a fresh variation.

T3Q05 · C

The intersection is C, and C→BD makes C determine the BCD component. The split is lossless. Projected rules provide C→B,D, but AB does not obtain C using them, so AB→CD is not preserved. Correct combination: lossless but not preserving.

Repair if missed: review R20, R22 and redo its worked example before trying a fresh variation.

T3Q06 · C

Each of four A entities has exactly one B, giving at most four distinct covered B entities. Total participation requires all five B entities to be covered. That is impossible. A product-size maximum does not remove the many-to-one restriction.

Repair if missed: review R24 and redo its worked example before trying a fresh variation.

T3Q07 · 6

The overflow leaf is [2,4,6]. The specified split is [2,4] | [6]. Copy the minimum of the right leaf,6, into the new root. A one-left/two-right split would have a different answer but is not the supplied convention.

Repair if missed: review R31 and redo its worked example before trying a fresh variation.

T3Q08 · 12

There are eight grouping sets. Four include the first dimension and have 2 groups each. Four omit it and have 1 group each, since the remaining dimension values are constant. Total=4×2+4×1=12. Eight grouping sets are not eight result rows.

Repair if missed: review R35 and redo its worked example before trying a fresh variation.

T3Q09 · 11

Population shares are 0.75 and 0.25. Weighted mean=0.75×8+0.25×20=6+5=11. The equal-stratum mean 14 ignores the population sizes.

Repair if missed: review R39 and redo its worked example before trying a fresh variation.

T3Q10 · D

Elapsed duration has a meaningful zero and ratios:20 seconds is twice10 seconds. Postal codes are nominal. Satisfaction categories only guarantee order. Celsius has an arbitrary zero, so a ratio of Celsius values is not the temperature ratio.

Repair if missed: review R37 and redo its worked example before trying a fresh variation.

Test 4 · Answer summary

Question	Answer
T4Q01	16
T4Q02	2
T4Q03	4
T4Q04	A; C
T4Q05	A
T4Q06	2
T4Q07	4
T4Q08	13
T4Q09	75
T4Q10	B; D

T4Q01 · 16

The unique key makes three attributes compulsory in each superkey. Each of the other four can be included or omitted. Count=2^4=16. The uniqueness of the candidate-key set is needed for this direct formula.

Repair if missed: review R17 and redo its worked example before trying a fresh variation.

T4Q02 · 2

P=1 has two witnesses and P=3 has one. P=2 only has S=0 and fails >0. Distinct qualifying X rows are 1 and 3:2 outputs, not3 witness pairs.

Repair if missed: review R08 and redo its worked example before trying a fresh variation.

T4Q03 · 4

Each left employee in team A matches all four right employees in A, including itself. Each employee in B matches two. Grouping by the left ID produces counts 4,4,4,4,2,2. Maximum:4.

Repair if missed: review R11, R14 and redo its worked example before trying a fresh variation.

T4Q04 · A; C

Keys are AB and AC; D is nonprime. A→D is a partial dependency on a proper subset of either key, so 2NF fails while atomic 1NF holds: A is true. It also violates BCNF, so B is false. AC is minimal and determines B,D, so C is true. C+ contains only B,C and misses A,D, so D is false.

Repair if missed: review R16, R19 and redo its worked example before trying a fresh variation.

T4Q05 · A

Local rules A→B,B→C,C→D preserve all original FDs. For losslessness, first split ABCD into ABC and CD: common C determines CD. Then split ABC into AB and BC: common B determines BC. Both steps are lossless. Thus the final three-way split is lossless and preserving.

Repair if missed: review R21, R22 and redo its worked example before trying a fresh variation.

T4Q06 · 2

Total one-to-one participation pairs every entity with exactly one on the other side. Merge the single-valued attributes into one relation with both original identifiers as candidate keys; their determinants are superkeys, so it is BCNF. Keep the multivalued attribute in a second owner-key/value relation. A single relation would repeat descriptive attributes across independent values and generally violate the stated normal-form target. Minimum:2. This relies on the expressly allowed merge.

Repair if missed: review R25, R26 and redo its worked example before trying a fresh variation.

T4Q07 · 4

A covering lookup reads one page at each of four uncached levels. All requested values are in the leaf. No separate data-page read is required:4.

Repair if missed: review R32 and redo its worked example before trying a fresh variation.

T4Q08 · 13

Overall sum=60+70=130 and count=3+7=10. Mean=13. The subgroup means are20 and10, but they have unequal weights; averaging them to15 is wrong.

Repair if missed: review R36 and redo its worked example before trying a fresh variation.

T4Q09 · 75

Position in the original interval=(25−10)/(30−10)=15/20=0.75. Map to [0,100]:0+100×0.75=75.

Repair if missed: review R37 and redo its worked example before trying a fresh variation.

T4Q10 · B; D

A is false: injective exact codes cannot shorten every input. B is true because the decoder needs the stored representation including its header. C is false: mean smoothing merges different inputs, generally losing exact values. D is true: short runs can cost more symbol/count bits than their original symbols. The alternating eight-character example has eight runs and expands.

Repair if missed: review R38, R40 and redo its worked example before trying a fresh variation.